Welding Precautions — Steel

Ford Ecat Electronic Parts Catalog

Material

Item	Specification
Motorcraft Premium Undercoating PM-25-A	—
Motorcraft Rust Inhibitor Aerosol PM-24-A	—

General Specifications - Welding Specifications

Item	Specification
Plug Weld Hole	8 mm (0.31 in)
Weld Wire ER70S-3 or equivalent	0.9-1.1 mm (0.035-0.045 in)

General Equipment
3 Phase Inverter Spot Welder 254-00002
Compuspot 700F Welder 190-50080
I4 Inverter Spot Welder 254-00014
Inverter Welder with MIG Welder 254-00015

WARNING: Invisible ultraviolet and infrared rays emitted in welding can injure unprotected eyes and skin. Always use protection such as a welder's helmet with dark-colored filter lenses of the correct density. Electric welding will produce intense radiation, therefore, filter plate lenses of the deepest shade providing adequate visibility are recommended. It is strongly recommended that persons working in the weld area wear flash safety goggles. Also wear protective clothing. Failure to follow these instructions may result in serious personal injury.

WARNING: Always wear protective equipment including eye protection with side shields, and a dust mask when sanding or grinding. Failure to follow these instructions may result in serious personal injury.

WARNING: On vehicles equipped with Safety Canopy® options, prior to carrying out any sectioning repairs near the roof line or sail panel areas of the vehicle, remove the Safety Canopy® module and related components. Failure to

comply may result in accidental deployment or damage to the Safety Canopy®. Refer to Section 501-20B. Failure to follow these instructions may result in serious injury to technician or vehicle occupant(s).

NOTICE: Electronic modules and related wiring can be damaged when exposed to heat from welding procedures. Carefully disconnect and remove, or position away from heat affected areas.

NOTE: *When it is necessary to carry out weld-bonding procedures, refer to Weld-Bonding in this section.*

The correct equipment and settings must be used when welding mild or High-Strength Steel (HSS) . Metal Inert Gas (MIG) and Squeeze-Type Resistance Spot Welding (STRW) are the preferred methods. Surfaces must be clean and free of foreign materials.

- Adequate ventilation must be provided to avoid accumulation of poisonous gases.
- A test weld should always be carried out on a test sample. Refer to the Weld Nugget Chart in Specifications for Ford-approved weld nugget information.
- Use cleaning brushes and abrasive grinding wheels dedicated to the type of materials being welded.
- Follow the equipment manufacturer's prescribed procedures and equipment settings for the type of welder being used. ER70S-3 or ER70S-6 wire are typically used for MIG welding steel.
- Disconnect the battery ground cable. Refer to Section 414-01.
- Disconnect on-vehicle modules adjacent to the welding area and protect them from possible heat damage and electrical currents when welding.
- Corrosion protection must be restored whenever bare metal repairs are made. Refer to Restoring Corrosion Protection Following Repair in this section.
- Adequate power supply needs to be used to make sure of correct equipment performance.
- Factory spot welds may be substituted with either STRW welds or MIG plug welds. Spot/plug welds should equal factory welds in both location and quantity. Do not place a new spot weld directly over an original weld location. Plug weld hole should equal 8 mm (0.31 in) diameter.

- Correct eye protection must be worn.
- The correct protective clothing should always be worn.
- Components made of HSS should not be heated to straighten or repair. If components are severely bent or kinked, new components should be installed.

Arc welding is an acceptable method for welding heavier metal components such as frame parts. When arc welding, the following guidelines should be followed:

- A temperature indicating crayon calibrated to 650°C (1,200°F) or below, should be used to avoid overheating and weakening the metal.
- Attach the ground clamp as close as possible to the work area.
- Choose welding electrodes according to the type of steel, thickness and polarity of the arc welder AC or DC.
- Do not use water or compressed air to cool welding. This can cause the metal to become brittle and weak.